

Multiple Choice

1. **Answer:** D

Options: A) All the numbers in the pattern can be divided equally by 10.; B) All the numbers in the pattern can be divided equally by 4.; C) All the numbers in the pattern can be divided equally by 8.; D) All the numbers in the pattern can be divided equally by 7.

Note: Correct option: D - All the numbers in the pattern can be divided equally by 7.

2. **Answer:** B

Options: A) -37; B) -9; C) 9; D) 37

Note: Correct option: B - -9

3. **Answer:** C

Options: A) 2 days; B) 4 days; C) 8 days; D) 9 days

Note: Correct option: C - 8 days

4. **Answer:** A

Options: A) 6-Jan; B) 5-Jan; C) 3-Jan; D) 2-Jan

Note: Correct option: A - 6-Jan

5. **Answer:** A

Options: A) 2; B) 4; C) 0; D) 1

Note: Correct option: A - 2

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '309'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(0, 9)'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-85'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** $y - 4 = \ln 2(x - 5)$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The probability of getting a test statistic at least as extreme as the observed test statistic, if the null hypothesis is true.. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key